LAB 5: Sequence Diagram – Library Management System

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Aim

To draw the sequence diagram of the Library Management System for the Issue Book scenario, showing how the objects interact with each other in time order.

Theory

A sequence diagram is an interaction diagram that shows how objects communicate with each other through messages, arranged in the order in which they happen. Time flows from top to bottom.

The elements used in the diagram are:

- **Actor / Object:** a participant of the interaction, drawn as a box at the top.
- **Lifeline:** the dashed vertical line below a participant that shows its existence over time.
- **Message:** a horizontal arrow from one lifeline to another, labelled with the operation that is called. A solid arrow is a call.
- **Return message:** a dashed arrow that sends the result back to the caller.
- **Combined fragment (alt):** a box that shows alternative paths. Only the path whose condition is true is executed, and the paths are separated by a dashed line.

Scenario

The scenario drawn is Issue Book. The librarian enters the member ID and the ISBN of the book. The system checks whether the member is eligible and whether the book is available. If both are true, an issue record is created, the stock of the book is reduced and a confirmation is shown to the librarian. Otherwise the request is rejected.

Objects Used

Object	Role
Librarian	The actor who starts the issue request.
:LibrarySystem	The control object that receives the request and coordinates the other objects.
:Member	Holds the details of the member and tells whether the member is eligible.
:Book	Holds the details of the book and its available copies.

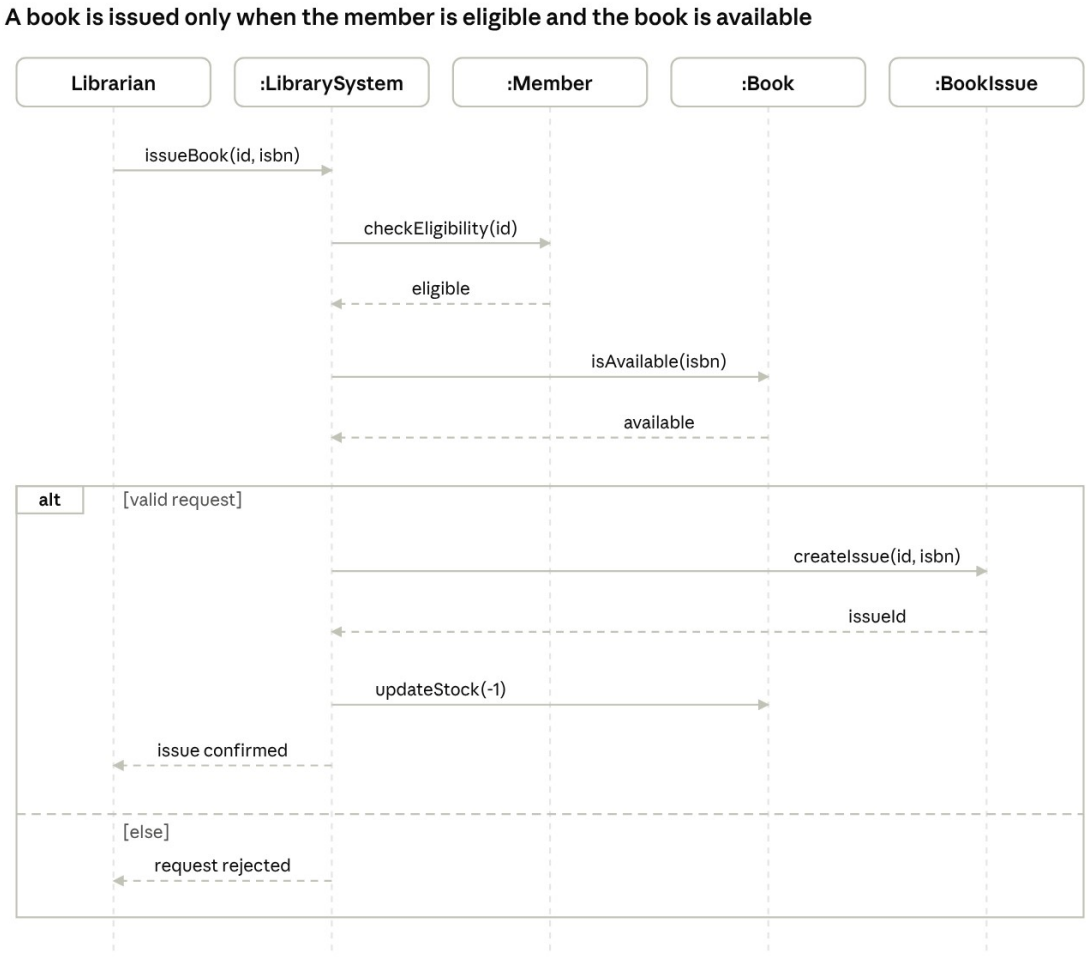
:BookIssue	The record that is created when a book is issued.

Message Sequence

No.	From	To	Message	Description
1	Librarian	:LibrarySystem	issueBook(id, isbn)	The librarian sends the member ID and the book ISBN.
2	:LibrarySystem	:Member	checkEligibility(id)	The system checks the pending fine and the issue limit of the member.
3	:Member	:LibrarySystem	eligible	Return message with the eligibility result.
4	:LibrarySystem	:Book	isAvailable(isbn)	The system checks whether a copy of the book is available.
5	:Book	:LibrarySystem	available	Return message with the availability result.
6	:LibrarySystem	:BookIssue	createIssue(id, isbn)	Valid request: the issue record is created with the issue date and due date.
7	:BookIssue	:LibrarySystem	issueId	Return message with the new issue ID.
8	:LibrarySystem	:Book	updateStock(-1)	Valid request: the available copies of the book are reduced by one.
9	:LibrarySystem	Librarian	issue confirmed	Valid request: the confirmation is shown to the librarian.
10	:LibrarySystem	Librarian	request rejected	Otherwise: the request is rejected and the librarian is informed.

Sequence Diagram

The sequence diagram of the Issue Book scenario is shown below.



sequence diagram · 5 lifelines, 1 alt fragment

Figure 1: Sequence diagram of the Issue Book scenario in the Library Management System.

Conclusion

The sequence diagram of the Issue Book scenario of the Library Management System was drawn. It shows the order of the messages between the Librarian, LibrarySystem, Member, Book and BookIssue objects, and the alternative path for a request that is not valid.